

4.5 Fact 5: Labor market adjustments are visible in employment more than compensation

In addition to employment we observe workers’ annual base compensation. We use this information to test for labor market adjustment along the compensation margin.¹⁹ Salary data are deflated to 2017 dollars using the PCE index.²⁰

Results are in Figure 10. The findings indicate a less marked divergence in compensation compared to employment across more and less exposed occupations. Figure 11 shows results by age and Eloundou et al. (2024)-based exposure quintile. We find little difference in compensation trends by age or exposure quintile.

Prior work by Autor and Thompson (2025) notes that technology that replaces inexpert tasks may reduce occupational employment but increase occupational wages; technology that replaces expert tasks may do the opposite. The sign of the wage effect depends on the overall share of tasks displaced as well as the whether these tasks are expert or inexpert. The limited changes we find for wages suggest that these effects may be offsetting, at least in the short run. Alternatively, the results could be explained by wage stickiness in the short run, consistent with recent evidence from Davis and Krolikowski (2025).

4.6 Fact 6: Findings are largely consistent under alternative sample constructions

We test the robustness of these results to alternative sample constructions and robustness checks.

Excluding Technology Occupations One possibility is that our results are explained by a general slowdown in technology hiring from 2022 to 2023 as firms recovered from the COVID-19 Pandemic.²¹ Figure A4 shows employment changes by age and exposure quintile after excluding computer occupations, corresponding to 2010 SOC codes that start with 15-1. Figure A5 shows results when excluding firms in the information sector (NAICS code 51). Results are quite similar,

¹⁹Total compensation may additionally include bonuses, overtime pay, commissions, equity, tips, and other items. These may have a greater impact on overall compensation in certain professions and age groups than others.

²⁰In contrast to the series for employment, results for compensation end in June 2025, the most recently available month for the PCE index.

²¹Under the Tax Cuts and Jobs Act, amendments to Internal Revenue Code §174 enacted in 2022 also disallowed companies from immediately deducting R&D expenditures, including software development costs. These costs instead had to be capitalized and amortized over five years for domestic research and fifteen years for foreign research.

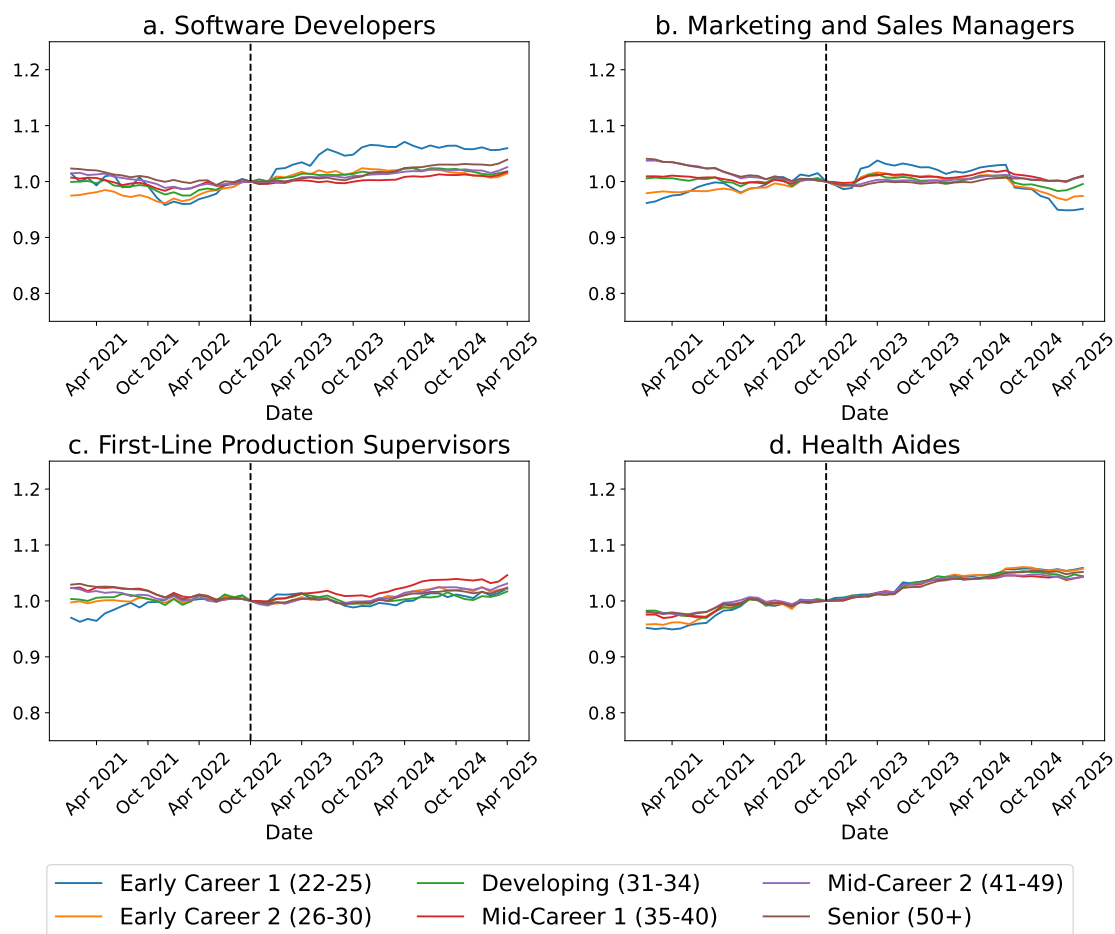


Figure 10: Changes in annual base compensation by age and occupation. Annual base compensation is deflated to 2017 dollars using the PCE deflator.

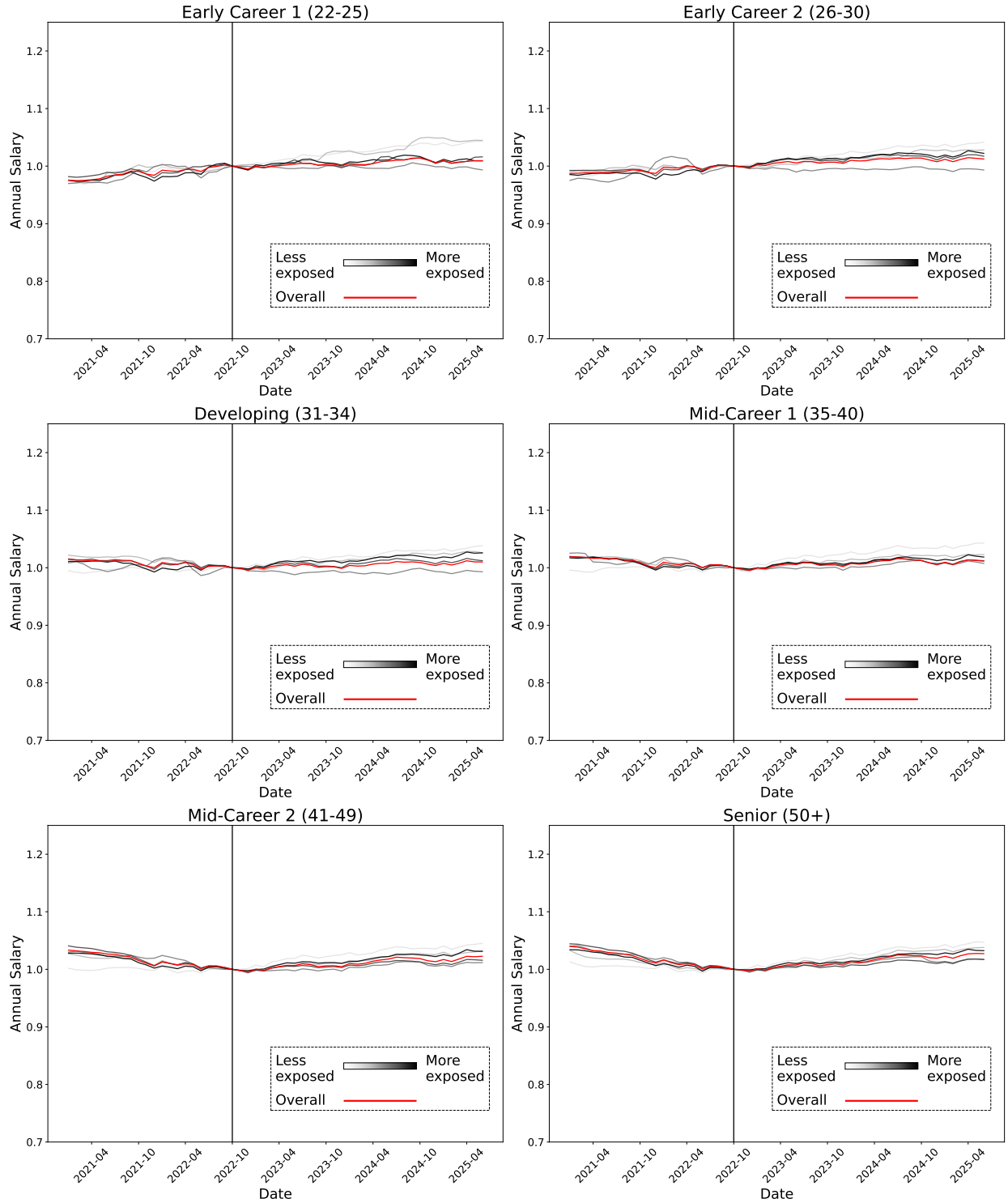


Figure 11: *Annual base compensation*. Changes in annual base compensation by age and exposure quintile. Exposure quintiles are defined based on the GPT-4 β measures from [Eloundou et al. \(2024\)](#). Darker lines are more exposed quintiles. The red line shows the overall trend pooling across quintiles. Annual base compensation is deflated to 2017 dollars using the PCE deflator.

consistent with the case studies above that show employment changes are visible across a range of occupations. Results with firm-time fixed effects in Figure 9 further show that our findings are robust to firm- or industry-level shocks that impact general hiring trends. These results indicate that our findings are not specific to technology roles.

Remote Work Figures A6 and A7 show results for occupations amenable to remote work (telework) and those that are not, according to Dingel and Neiman (2020).²² We find that, for young workers, more exposed occupations have slower employment growth, both in teleworkable occupations and in non-teleworkable occupations. The results for non-teleworkable occupations in particular suggest that our findings are not driven by outsourcing or work-from-home disruptions, at least solely.²³

Longer Sample Figure A8 shows results when extending the balanced sample of firms to 2018. This reduces the sample size and makes the data somewhat noisier. Nonetheless, the trends remain largely ordered by exposure in the post-GPT era, whereas this is not the case before 2022. A concern is that for the Eloundou et al. (2024) measures the most exposed quintile had slower employment growth starting around 2020. This is not the case for the Anthropic exposure measures, shown in Figures A9, A10, and A11. For these measures the most exposed groups have comparable employment growth throughout the period before generative AI, with divergent trends afterwards.

Changes in Education Another possibility is that the changes we observe are influenced by worsening education outcomes during the COVID-19 Pandemic. COVID-19 led to persistent harms in education outcomes (Kuhfeld and Lewis, 2025). As more educated workers on average have greater measured AI exposure, deterioration in the quality of education in recent years could influence the trends we observe.²⁴ In Figure A12 we show trends for occupations in which greater than

²²Whether an occupation is amenable to remote work is positively correlated with AI exposure. Only two teleworkable occupations fall in the lowest quintile of estimated AI exposure according to the GPT-4 β measure. Likewise few non-teleworkable occupations fall in the highest quintile of AI exposure. For this reason in Figure A6 we pool together the two lowest AI exposure quintiles into one group. In Figure A7 we pool together the two highest AI exposure quintiles.

²³Non-teleworkable occupations with high AI exposure include bank tellers, travel agents, and tax preparers.

²⁴Chandar (2025a) finds that declines in average skill levels for college graduates explain a sizable share of the slowdown in the growth of the college wage gap in recent decades.